resolution is projective (where we agree that $P_{-1} = M$). Thus, by replacing P_d with K_d and all terms to the left of P_d by zero, we get a projective resolution of M of length d. Deduce that if A and M are finite dimensional, then there is a finite resolution $P_\bullet$ of M with finite dimensional P_i.

Definition 9.4.3. One says that a ring A has left (respectively, right) **homological dimension** $\leq d$ if every left (respectively, right) A-module M has projective dimension $\leq d$. The homological dimension of A is exactly d if it is $\leq d$ but not $\leq d-1$. If such a d does not exist, one says that A has infinite homological dimension.

Example 9.4.4. By the Hilbert syzygies theorem (see Problem 8.2.10(iv)), the homological dimension of the polynomial algebra $k[x_1, \ldots, x_n]$ is n.

Problem 9.4.5. (i) Show that if a finite dimensional algebra A has finite homological dimension d and if C is the Cartan matrix of A, then $\det(C) = \pm 1$.

(ii) What is the homological dimension of $k[t]/t^n$, $n > 1$? Of the algebra of Problem 9.3.2?

Problem 9.4.6. (i) Show that the path algebra P_Q of any quiver Q with at least one edge has homological dimension 1. In particular, the homological dimension of the free algebra $k\langle x_1, \ldots, x_n \rangle$ is 1 (for $n \geq 1$).

(ii) Let Q be a finite oriented graph without oriented cycles. Find the Cartan matrix of its path algebra P_Q.

9.5. Blocks

Let A be a finite dimensional algebra over an algebraically closed field k, and let $\mathcal{C}$ denote the category of finite dimensional A-modules.

Definition 9.5.1. Two simple finite dimensional A-modules X, Y are said to be **linked** if there is a chain $X = M_0, M_1, \ldots, M_n = Y$ such that for each $i = 0, \ldots, n-1$ either $\mathrm{Ext}^1(M_i, M_{i+1}) \neq 0$ or $\mathrm{Ext}^1(M_{i+1}, M_i) \neq 0$ (or both).